

Methodology — Neuromorphic Sleep Stage Scoring

Research Approach

This project follows a systematic engineering methodology for developing an automated sleep-stage classification system. The approach emphasizes reproducibility, edge-deployment readiness, and exhibition-quality documentation.

Pipeline Overview

METHODOLOGY PIPELINE

Phase 1: Data Acquisition & Governance

- └─ Sleep-EDF download & validation
- └─ PSG/Hypnogram pairing
- └─ Subject-level splitting
- └─ Manifest generation

Phase 2: Signal Preprocessing

- └─ Bandpass filtering (0.5-35 Hz)
- └─ Notch filtering (50 Hz)
- └─ Z-score normalization
- └─ Artifact quality control
- └─ Epoch caching

Phase 3: Exploratory Data Analysis

- └─ Class distribution analysis
- └─ Signal characteristic visualization
- └─ Temporal pattern analysis
- └─ Quality control validation

Phase 4: Model Development

- └─ Architecture design (Improved Student)
- └─ Teacher training (Improved Teacher)
- └─ Knowledge distillation
- └─ Checkpoint selection

Phase 5: LoRA Adaptation

- └─ Rank sweep ($r=2$, $r=4$, $r=8$)
- └─ 4-fold held-out-subject CV
- └─ Multi-seed confirmation
- └─ Latency & merge verification
- └─ Parameter-efficiency analysis

Phase 6: Evaluation & Deployment

- └─ Test set evaluation
- └─ Official result documentation
- └─ Edge deployment preparation
- └─ Exhibition demonstration

Phase 1: Data Acquisition & Governance

Objectives

- Obtain Sleep-EDF PSG recordings
- Validate data integrity
- Create reproducible subject-level splits
- Establish data contract for downstream processing

Methods

Data Source: PhysioNet Sleep-EDF Expanded database

- 78 healthy subjects
- 2 nights per subject (most subjects)
- 100 Hz sampling rate
- 4 channels: EEG Fpz-Cz, EEG Pz-Oz, EOG, EMG

Subject-Level Splitting:

```
# Prevents data leakage from same subject in multiple splits
train_subj, temp_subj = train_test_split(subjects, test_size=0.30, random_state=42)
val_subj, test_subj = train_test_split(temp_subj, test_size=0.50, random_state=42)
```

Quality Assurance:

- PSG-Hypnogram file pairing validation
- Duplicate subject-night detection
- Channel availability verification

Deliverables

- data/manifests/sleep_edf.csv — Subject/night manifest
- data/manifests/subject_splits.csv — Train/val/test splits

Phase 2: Signal Preprocessing

Objectives

- Reduce noise and artifacts
- Standardize signal characteristics

- Create model-ready epoch representations

Methods

Bandpass Filtering:

- 4th-order Butterworth filter
- Cutoff: 0.5 Hz (high-pass) to 35 Hz (low-pass)
- Rationale: Removes baseline drift and high-frequency noise while preserving sleep-relevant frequencies

Notch Filtering:

- IIR notch filter at 50 Hz
- Quality factor: 30
- Rationale: Eliminates power-line interference

Z-Score Normalization:

```
x_normalized = (x - mean) / std
```

- Per-channel, per-epoch
- Ensures zero mean and unit variance
- Makes model invariant to absolute signal amplitude

Artifact Quality Control: | Check | Threshold | Action | |-----|-----|-----| | Clipping | $|x| > 8.0\sigma$ | Flag epoch | |
Flatline | $\text{std} < 0.05$ | Flag epoch | | NaN/Inf | Non-finite values | Flag epoch |

Rationale

Preprocessing is treated as a first-class engineering step, not an optional cleanup. The contract is fixed and reproducible.

Deliverables

- `data/cache/*.npz` — Preprocessed epoch arrays
- `data/cache/cache_index.csv` — Cache metadata

Phase 3: Exploratory Data Analysis

Objectives

- Verify dataset characteristics
- Identify class imbalance
- Validate preprocessing quality
- Understand temporal patterns

Analyses Performed

Class Distribution:

- Quantified imbalance across 5 sleep stages
- N2 dominates (~50% of epochs)
- N1 and REM are minority classes

Signal Characteristics:

- Visualized raw vs. preprocessed waveforms
- Verified frequency content after filtering
- Confirmed normalization effectiveness

Temporal Patterns:

- Analyzed sleep-stage transition frequencies
- Verified contiguous epoch ordering
- Confirmed no data leakage across recordings

Key Findings

- Class imbalance requires imbalance-aware training (focal loss, class weights)
- Artifact flag rate ~2% (acceptable)
- Strong temporal dependencies justify sequence modeling

Phase 4: Model Development

Architecture Design Principles

- Multi-scale signal processing:** Capture patterns at different temporal scales
- Computational efficiency:** Use depthwise-separable convolutions for edge deployment
- Frequency awareness:** Include parametric Gabor filters for spectral patterns
- Temporal modeling:** Use GRU to capture stage transitions

Improved Student Architecture

Component	Description	Parameters
Multi-Resolution Stem	Parallel short/long receptive fields	~1,000
Depthwise-Separable CNN	Efficient feature extraction	~2,000
Parametric Gabor FEB	Learnable frequency filters	~1,300
2-Layer GRU	Temporal sequence modeling	~16,500
Classification Head	5-class softmax	320
Total		99,477

Training Strategy

Knowledge Distillation:

- Train a larger "teacher" model first
- Distill knowledge into the smaller "student"
- Student learns from both hard labels and softened teacher predictions

Loss Function:

$$L_{total} = \alpha_{ce} \times L_{CE} + \alpha_{kl} \times L_{KL} + \alpha_{feat} \times L_{Feature}$$

Component	Weight	Purpose
Cross-entropy	1.0	Hard-label supervision
KL divergence	1.0	Teacher-student alignment
Feature MSE	0.5	Intermediate representation alignment

Hyperparameters:

Parameter	Value	Justification
Optimizer	AdamW	Adaptive learning rate with weight decay
Learning rate	3e-4	Standard for small models
Batch size	16	Balance between speed and stability
Sequence length	10	5 minutes of context (300s)
Training epochs	20	Sufficient for convergence

Rationale

The architecture is intentionally compact for edge deployment. The 99,477 parameter count is below the project's 100K target while maintaining competitive accuracy (87.5% across 15 subjects).

Deliverables

- artifacts/student_improved_best.pt — Final trained model
- artifacts/teacher_improved_best.pt — Teacher model (training only)

Phase 5: Evaluation & Deployment

Evaluation Protocol

Test Set: Held-out subjects never seen during training

Metrics:

Metric	Value	Interpretation
Accuracy	87.5% $\pm$ 3.2%	Overall correct classifications
Cohen's κ	0.763 $\pm$ 0.043	Agreement beyond chance
Macro F1	0.721 $\pm$ 0.050	Balanced performance across classes

Per-Class Performance:

Stage	F1 Score	Notes
Wake	0.9693	Excellent — strong EEG/EMG signatures
N1	0.2006	Challenging — brief, transitional stage
N2	0.8162	Good — distinct spindle/K-complex features
N3	0.7849	Good — clear delta wave patterns
REM	0.3586	Moderate — overlap with Wake/N1

Official Result

The project's single authoritative result is:

```
Improved Student
87.34% Test Accuracy
Cohen's  $\kappa$  = 0.7551
99,477 Parameters
8.5 ms/batch CPU latency
```

Deployment Preparation

1. Export to ONNX format
2. Apply INT8 post-training quantization
3. Validate on target hardware
4. Implement real-time inference pipeline

Reproducibility

Aspect	Implementation
Random seeds	Python, NumPy, PyTorch all seeded to 42
Deterministic operations	CUDA deterministic mode enabled
Configuration	YAML files for all parameters
Version control	Git-tracked code and manifests
Checkpointing	Best model saved by validation κ

Ethical Considerations

1. **Not a medical device:** This is a research prototype, not a clinical diagnostic system
2. **Limited validation:** Only validated on Sleep-EDF (healthy subjects)
3. **Expert oversight required:** Should supplement, not replace, expert scoring
4. **Data privacy:** Uses publicly available, de-identified data
5. **Responsible claims:** Avoid overstating clinical applicability

Last updated: August 2026 Project: Neuromorphic Sleep Stage Scoring — VIT Bhopal University

